

# Collections Copilot

## Helping collections employees prepare a case and respond to customers

A unified customer view, relevant policy and suggested assistance

Team 5C2. Project overview for rubric discussion.

# Four connected layers

- 1 Data foundation: match identities across five systems, validate records and build a customer C360 with explicit missingness.
- 2 Questions: answer supported collections questions using trusted SQL, current policy or call evidence, with sources and refusals.
- 3 Features: serve 12 versioned case features with the same offline and online definitions, plus text signals.
- 4 Employee assistance: suggest a response or next step, explain its evidence and record accept, edit or reject decisions.

# Six use cases

## Agent Assist

Help during the conversation, including hardship, disputes and policy guidance.

## Next Best Action

Explain action reasons, alternatives and programme eligibility prerequisites.

## Channel comparison

Show outcome support, uncertainty, timing and unmet operational checks.

## Agent routing

Match all required skills and language; explain capacity and roster exclusions.

## Post-call summary

Separate proposed terms, explicit commitments and follow-ups with timestamps.

## Quality assurance

Review context for ten checklist items and record employee evidence per item.

Local reviewed workflows. Treatment effectiveness and production readiness remain unverified.

# AI and model choices

Team-trained model: a small text classifier trained on approved notes and customer transcript text to identify financial hardship.

Local models: Whisper transcribes audio. A team-trained Qwen LoRA selects cited case facts. Both run locally without paid API calls.

Optional paid API: an adapter is implemented, but needs user credentials and cost approval. It remains untested.

Controls: deterministic contact gates and employee review remain in place for every model approach.

# Current development evidence

114 automated checks pass. The proposal safeguard audit checks 719 approved cases with 0 failures of tested constraints.

All 12 numeric development benchmarks and 24 source-grounded policy/scope checks pass. Broad semantic accuracy remains unmeasured.

Hardship, 181 development examples: classifier 98.6% precision/recall. Live combination: 94.7% precision, 100% recall.

Whisper small.en: 11.02% word error rate on 1803 approved synthetic recordings. Speaker review remains required.

LoRA: 1529 training examples. 468 final factual-format outputs, 100.0% complete exact-value results. Narrow synthetic task only.

Hidden-test results and preparation-time savings remain unmeasured.

# Possible grading dimensions

Data correctness: matching quality, financial totals, quarantine, coverage and a reproducible C360 contract.

Question accuracy: benchmark tolerance, usable SQL or source evidence, policy applicability and correct refusals.

Model quality: separate training and evaluation, precision/recall, transcription error and factual draft quality.

Workflow quality: depth of the chosen use case, usability, response time and useful explanations.

Safeguards and reproducibility: consent, holds, privacy, employee control, executable code and honest limitations.

Organiser decisions: paid APIs, pretrained models, use-case breadth and scoring weights.